7

SCIENTIFIC NOTATION AND PROBLEM SOLVING INVOLVING REAL NUMBERS

Learner's Module in Mathematics 7

Quarter 1 • Module 9

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PREFACE

This module is a project of the Curriculum Implementation Division particularly the Learning Resource Management and Development Unit, Department of Education, Schools Division of Baguio which is in response to the implementation of the K to 12 Curriculum.

This learning Material is a property of the Department of Education- CID, Schools Division of Baguio. It aims to improve students' performance specifically in Mathematics.

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Learning Competencies/codes:	Writes numbers in scientific notation and vice versa. **M7NS-Ii-1** Represents real-life situations and solves problems involving real numbers.

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WHAT I NEED TO KNOW

Welcome to this module on SCIENTIFIC NOTATION AND PROBLEM SOLVING INVOLVING REAL NUMBERS!

This module is intended to help you, learners, understand and master the concepts of scientific notation. It is designed to equip you with essential knowledge about the said topic and skills on solving problems involving real numbers.

HOW TO USE THIS MODULE

This module contains several lessons. To make the most out of them, you need to do the following:

1. Scan the list of *Learning Objectives* to get an idea of the knowledge and skills you are expected to gain and develop as you study the module. These outcomes are based on the content standards, performance standards, and learning competencies of the K to 12 Curriculum Mathematics 7.
2. Take the *What I Know*. Your score will determine your knowledge of the lessons in the module. If you get 100% of the items correctly, you may proceed to the next module. Otherwise, you must go through the lessons.
3. Each *Lesson* aims to develop one of the learning objectives set for the module. It starts with an activity that will help you understand the lesson and meet the required competencies.
4. Take your time in reading this module. Write down points for clarification. You may discuss these points with your teacher.
5. Perform all activities and answer all worksheets. The activities and assessments are designed to enhance your understanding of scientific notation and application on real numbers. The activities will also give you an idea how well you understand the lessons. Review the lessons if necessary, until you have achieved a satisfactory level of understanding.
6. At the end of the module, take the *Post-Assessment* to evaluate your overall understanding about the lessons.

LEARNING OBJECTIVES

At the end of the module, you should be able to:

1) determine the significant digits in a given situation.
2) write numbers in scientific notation and vice versa.
3) solve problems involving real numbers.

DAY 1 WHAT I KNOW

Instruction: Let us determine how much you already know about scientific notation and problems involving real numbers. Read each item carefully. Provide the correct answer by writing it on your answer sheet.

1) How many significant digits are there in 2019?

a) 1 b) 2 c) 3 d) 4

2) Which number has 4 significant digits?

a) 0.0018 b) 0.0001 c) 8,000 d) 2002

3) Which value shows a significant zero - digit?

a) 0.01 b) 0.002 c) 3.04 d) 500

4) How many significant digits are there in 0.0123?

a) 1 b) 2 c) 3 d) 4

5) How many significant digits should the product of 1.2 x 3.7?

a) 1 b) 2 c) 3 d) 4

6) Why is scientific notation important to real numbers?

a) Because it makes real numbers mathematical
b) Because it helps in mathematical computations
c) Because it simplifies big and small numbers
d) Because it makes use of the powers of 10

7) In the scientific notation 3.0×10^2, which is the coefficient?

a) 2 b) 3 c) 10 d) x

8) In the scientific notation 1.23×10^{-2}, which shows the number of times the decimal place is moved?

a) – 2 b) 10 c) 0.23 d) 1.2

9) Which part of the scientific notation indicates the direction of the movement of the decimal point?

a) Power of 10
b) Sign of coefficient
c) Sign of exponent
d) Position of decimal point

10) Which is the scientific notation of 789,000,000,000?

a) 789×10^9
b) 78.9×10^{10}
c) 7.89×10^{11}
d) 0.789×10^{12}

11) Which is the scientific notation of 0.000789?

a) 789×10^{-4}
b) 78.9×10^{-4}
c) 7.89×10^{-4}
d) 0.789×10^{-4}

12) Which is the standard form of the scientific notation 4.3×10^{5}?

a) 430,000 b) 43,000 c) 0.00043 d) 0.0043

13) Which is the standard form of the scientific notation 4.3×10^{-5}?

a) 0.000043 b) 0.00043 c) 0.0043 d) 0.043

14) When a number is written in scientific notation, where is the decimal point located?

a) Decimal point is located between the last two digits of the given number
b) Decimal point is located after the last non-zero digit of the given number.
c) Decimal point is located after the first non-zero digit of the given number.
d) Decimal point is located between any two non-zero digits of the number.

15) When the exponent of 10 in the scientific notation of a number is negative, it means that the number

a) is less than 1.
b) is greater than 1
c) is equal to 1.
d) could not be determined.

16) A deep-freeze compartment is maintained at a temperature of 12°C below zero. If the room temperature is 31°C, how much warmer is the room temperature than the temperature in the deep-freeze compartment.

a) 43°C b) 44°C c) 45°C d) 46°C

17) Yohan can type 50 words per minute, how many words can Yohan type in 20 minutes?

a) 10 words b) 100 words c) 1000 words d) 500 words

18) A hair dryer uses 1.2 kilowatt (KW) of power while a hair iron uses 1.14 kilowatt (KW) of power. How much power is needed if you wish to use the two at the same time?

a) 2.33 KW b) 2.34 KW c) 2.35 KW d) 2.36 KW

19) An elevator is on the ground floor. It goes up 8 floors, then 5 down, and the 4 up. What is its final position from the ground floor?

a) 4^{th} floor b) 5^{th} floor c) 6^{th} floor d) 7^{th} floor

20) Hanna donated $\frac{2}{5}$ of her monthly allowance to the typhoon victims. If her monthly allowance is 3,500 pesos, how much did she donate?

a)1000 pesos b) 1200 pesos c) 1400 pesos d) 1600 pesos

Lesson 9.1 SIGNIFICANT DIGITS

What's In?

We sometimes say that we have significant and insignificant experiences. These experiences depend on how it affects our daily life. Numbers too, can be considered significant or insignificant depending on the degree of precision made. How can we tell when a certain computation is significant to consider? What guidelines can we use to help us determine the significance of a computation? What then makes a computation significant? Do you want to know more about it? Read on!

What's New?

Examine the table below. The first column is the list of numbers and the second column is the number of significant digits.

List of Numbers	**Number of Significant Digits**
234	3
745.1	4
6007	4
0.012300	5
100.0	4
100	1
7890	3
4970.00	6
1.3×10^{2}	2
7.50×10^{-2}	3

Examine the relationship of the numbers and the number of significant digits in each.

What rule on significant digits can you conclude?

You may review your answer after going through the examples.

What's in it?

Definition:

Significant digits are digits that carry meaning according to their precision.
Note: The number of significant digits in a given measurement depends on the number of significant digits in the given data.

Rules for significant digits:

Rule 1: All nonzero digits are significant.

Example 1:

Nonzero Digits	Number of Significant Digits
8	1
118	3
3 459	4
56.342	5
1487.486	7

Rule 2: All zeros located between nonzero digits are significant digits.

Example 2: 108 has _3_ significant digits; 1, 0, and 8
2005 has _4_ significant digits; 2, 0, 0, and 5
4800006 has _7_ significant digits; 4, 8, 0, 0, 0, 0, and 6
30.65 has _4_ significant digits; 3, 0, 6, and 5
56.0015 has _6_ significant figures; 5, 6, 0, 0, 1, and 5

Rule 3: All zeros at the end of a decimal are significant

Example 3: 2.0 has _2_ significant digits; 2 and 0
15.0 has _3_ significant digits; 1, 5, and 0
21.30 has _4_ significant digits; 2, 1, 3, and 0
5.250 has _4_ significant digits; 5, 2, 5, and 0
78000.00 has _7_ significant digits; 7, 8, 0, 0, 0, 0, and 0

Rule 4: All digits of the first factor when a number is expressed in scientific notation are significant.

Example 4: 2×10^3 has _1_ significant digit; 2 only
8.4×10^2 has _2_ significant digits; 8 and 4
4.38×10^6 has _3_ significant digits; 4, 3, and 8
5.607×10^{-2} has _4_ significant digits; 5, 6, 0, and 7
6.0030×10^{-3} has _5_ significant digits; 6, 0, 0, 3 and 0

Rule 5: Underscored or specified zeros of a whole number ending in zeros are significant.
A decimal placed after the zeros are also significant.

Example 5: 180 has *3* significant digits; 1, 8, and 0
7000 has *4* significant digits; 7, 0, 0, and 0
89000 has *4* significant digits; 8, 9, 0, and 0
90. has *2* significant digits; 9 and 0
6300. has *4* significant digits; 6, 3, 0, and 0

Digits that are not significant
Rule 6: Zeros at the end of a whole number are not significant

Example 6: 90 has *1* significant digit; 9 only
5030 has *3* significant digits; 5, 0, and 3
230000 has *2* significant digits; 2 and 3
50000000 has *1* significant digit; 5 only
780004000 has *6* significant digits; 7, 8, 0, 0, 0, and 4

Rule 7: Zeros following the decimal point in a number between 0 and 1 are not significant.

Example 7: 0.05 has *1* significant digit; 5 only
0.0000009 has *1* significant digit; 9 only
0.012 has *2* significant digits; 1 and 2
0.00562 has *3* significant digits; 5, 6, and 2
0.00890 has *3* significant digits: 8, 9, and 0

What's More?

Note: *This is a guided practice. Try to answer first the activity before checking the solutions after it.*

Activity 1 – A : Complete the table below.

List of numbers	Number of Significant Digits	Indicate the rule number that was applied
1) 0.0122		
2) 0.00430		
3) 300668		
4) 10000		
5) 1000.		

List of numbers	Number of Significant Digits	Indicate the rule number that was applied
6) 2.333×10^{-3}		
7) 8.004×10^{5}		
8) 612$\underline{0}$		
9) 120.0		
10) 530		

*[**Answer:** Number of significant digits; 1) 3, 2) 3, 3) 6, 4) 1, 5) 4, 6) 4, 7) 4, 8) 4, 9) 4, 10) 2; Rule Number applied; 1) 7, 2) 7, 3) 2, 4) 6, 5) 5, 6) 4, 7) 4, 8) 5, 9) 3, 10) 6]*

Activity 1 – B : Four students weigh an item using different scales. These are the values they report:

a. 30.04 g
b. 30.0 g
c. 0.3004 kg
d. 30 g

How many significant digits are in each measurement?

*[**Answer:** a) 4, b) 3, c) 4, d) 1]*

Activity 1 – C : A hair dryer uses 1.2 kilowatt (KW) of power while a hair iron uses 1.14 kilowatt (KW) of power. How much power is needed if you wish to use the two at the same time?

*(**Answer:** 1.2 + 1.14 = 2.34 KW)*

Assessment 1

Three students measure volumes of water with three different devices. They report the following results:

Device	Volume
Large graduated cylinder	175 mL
Small graduated cylinder	39.7 mL
Calibrated buret	18.16 mL

If the students pour all of the water into a single container, what is the total volume of water in the container? Indicate the number of significant digits.

What I Have Learned?

Instruction: Complete the 3-2-1 table below.

3 things I learned	2 things I want to learn more	1 thing I mastered

Day 2

Lesson 9.2 SCIENTIFIC NOTATION

What's In?

Do you know that the earth weighs 6,588,000,000, 000,000,000,000,000 tons? Have you heard that the human body contains 70,000 miles of blood vessels? Do you know that it takes roughly 0.000000003 seconds for light to travel one meter? These are some examples of big and small numbers facts. Do you think it is possible to read and write these numbers in an easier way? Read on to find out!

What's New?

Try representing 6,588,000,000,000,000,000,000,000 tons. Write your idea in the box. You can use any mathematical concept or method you have previously studied to represent this very big number easily.

What's in it?

Scientific notation *simplifies the way we write very large and very small numbers in a compact form. The primary components of a number written in a scientific notation are as follows:*

$n \times 10^{e}$

Coefficient (number between 0 and 10)

integer *exponent*

Base of 10

WRITING NUMBER (STANDARD FORM) TO SCIENTIFIC NOTATION

A. For a large number

Steps:

a) Move the decimal point to the left until you have a number greater than or equal to 1 and less than 10.

b) Count the number of decimal places you moved the decimal point to the <u>*left*</u> and use that number as the <u>*positive power of 10.*</u>

c) Multiply the decimal (in step a) by the power of 10 (in step b).

Examples: 1. Write 782,000 in scientific notation

Solution:

Start by putting decimal point to the last digit then move the decimal to the left until it reaches the first significant digit which is number 7.

Illustration:

7. 8 2, 0 0 0.

5 4 3 2 1

Since the decimal point was moved 5 places to the left,

multiply by 10^5 ⟶ ***(number of moves of the decimal point)*** .

Answer: 782,000 = 7.82×10^5

(Sign of the exponent should be positive)

2. Write 7,000,000 in scientific notation.

Solution:

7. 0 0 0 0 0 0.

6 5 4 3 2 1

Answer: 7,000,000 = 7×10^6

3. Write 2,479.80 in scientific notation.

Solution:

2. 4 7 9. 8 0

3 2 1

Answer: 2,479.80 = 2.4798×10^3

4. Write 348,000.00 in scientific notation.

Solution:

3.4 8 0 0 0. 0 0

5 4 3 2 1

Answer: 348,000.00 = 3.48×10^{5}

B. For a small number

Steps:

a) Move the decimal point to the right until you have a number greater than or equal to 1 and less than 10.

b) Count the number of decimal places you moved the decimal point to the *right* and use that number as the *negative power of 10.*

c) Multiply the decimal (in step a) by the power of 10 (in step b).

Examples: 1. Write 0.005 to scientific notation.

Solution:

Move the decimal to the right until it reaches the first significant digit which is 5.

Illustration:

0 . 0 0 5.

1 2 3

Answer: 0.005 = 5×10^{-3}

(Sign of the exponent should be negative)

2. Write 0.000451 in scientific notation.

Illustration:

0 . 0 0 0 4.51

1 2 3 4

Answer: 0.000451 = 4.51×10^{-4}

What's More?

Activity 2

Write each number in scientific notation. Choose your answer from the box below.

1) 234,550
2) 660,790
3) 20,000
4) 2,900
5) 6,004,700
6) 0.0003605
7) 0.0003200
8) 0.006
9) 0.000078
10) 0.00012

1.2×10^{-4}	7.8×10^{-5}	2.3455×10^{5}	
6.6079×10^{5}	6×10^{-3}	6.0047×10^{6}	
3.605×10^{-4}	2×10^{4}	3.2×10^{-4}	2.9×10^{3}

*[**Answer:** 1)* 2.3455×10^{5}; 2) 6.6079×10^{5}; 3) 2×10^{4}; 4) 2.9×10^{3}; 5) 6.0047×10^{6}; 6) 3.605×10^{-4}; 7) 3.2×10^{-4}; 8) 6×10^{-3}; 9) 7.8×10^{-5}; 10) 1.2×10^{-4} *]*

Assessment 2

"A Greek Mathematician who first developed the concept of scientific notation" Do you know his name? Find out by decoding the hidden message below.

Match Column A with its answer in Column B to know the name of that Greek Mathematician. Put the letter of the correct answer in the space provided below.

Column A	Column B
1) 625	A) 6.25×10^{2}
2) 4216	C) 4.9603×10^{4}
3) 49,603	D) 9.12×10^{-3}
4) 250,000	E_1) 1.8×10^{-2}
5) 800,000,000	E_2) 1.3×10^{-5}
6) 0.025	H) 2.5×10^{5}
7) 0.01800	I) 8×10^{8}
8) 0. 00912	M) 2.5×10^{-2}
9) 0.000013	R) 4.216×10^{3}

Decode:

___ ___ ___ ___ ___ ___ ___ ___ ___ ___

1 2 3 4 5 6 7 8 9 10

What I Have Learned?

Instruction: To sum it up, let us complete the statement. Choose your answer from the box that best completes the statements below.

compact	coefficient	base 10
positive	negative	

1. Scientific notation allows us to express a very small or very large number in a ________________ form.
2. Scientific notation is composed of a number part called ___________ (a number between 0 and 10).
3. Scientific notation is composed of a number with _____________ raised to an integer power.
4. When the decimal is moved towards the left, the sign of the exponent should be _______________.
5. When the decimal is moved towards the right, the sign of the exponent should be _______________.

Day 3

Lesson 9.3 WRITING SCIENTIFIC NOTATION TO STANDARD FORM

What's In?

Recall the steps of writing number in scientific notation by analyzing the problem below.

Telephone Subscribers

In 2007, the total number of mobile telephone subscribers was estimated at 57,300,000, almost 15 times more than the number of fixed – line telephone subscribers which is estimated at 3,900,000. Write the numbers of mobile telephone and fixed – line telephone subscribers in scientific notations.

What's New?

Given the scientific notation 8.09×10^{10}, do you know how to expand the number to standard form?
The answer to the question will be discuss as we go further with discussion on rewriting scientific notation to standard form.

What's in it?

Writing Scientific Notation to Standard Notation

Steps:

a. To change scientific notation with a positive integer exponent to standard form, move the decimal point to the right, the number of places indicated by the exponent.
b. To write a number expressed in scientific notation with negative exponent in standard form, move the decimal point to the left, the same number of places as the absolute value of the exponent.

Examples: Write each number in standrad form.

1. 7×10^3

   Solution:

   Since the exponent is positive 3, then decimal point will be moved to the right 3 times.

   Illustration: 7• 0 0 0•

   1 2 3

   Answer: $7 \times 10^3 = 7{,}000$

2. 8.4×10^5

   Solution: 8 • 4 0 0 0 0 •

   1 2 3 4 5

   Answer: $8.4 \times 10^5 = 840{,}000$

3. 8.13×10^4
   Solution:

   8 •1 3 0 0•
   1 2 3 4

   Answer: $8.13 \times 10^4 =$ 81,300

4. 5×10^{-4}

   Solution:
   Since the exponent is negative 4, then the decimal point will be moved to the left 4 times.

   Illustration:

   •0 0 0 5 •
   4 3 2 1

   Answer: $5 \times 10^{-4} = 0.0005$

5. 6.49×10^{-6}

   Solution:

   •0 0 0 0 0 6• 4 9
   6 5 4 3 2 1

   Answer: $6.49 \times 10^{-6} = 0.00000649$

What's More?

Activity 3 – A

Write each number in standard notation.

1) 2.34×10^2
2) 1.567×10^5
3) 7×10^6
4) 5.9×10^9
5) 5.39×10^5
6) 2.34×10^{-2}
7) 1.567×10^{-5}
8) 7×10^{-6}
9) 5.9×10^{-4}
10) 5.39×10^{-5}

*[**Answer:** 1) 234; 2) 156,700; 3) 7,000,000; 4) 5,900,000,000; 5) 539,000; 6) 0.0234; 7) 0.00001567; 8) 0.000007; 9) 0.00059; 10) 0.0000539]*

Activity 3 – B
Analyze each problem then answer what is being asked.

1. The speed of light is 186,000 miles per second, or about 11,160,000 miles per hour. Express these numbers in scientific notation.

*(**Answer:** 186,000 = 1.86×10^5 miles per second; 11,160,000 = 1.116×10^7 miles per hour)*

2. The radius of a hydrogen atom is 2.5×10^{-11} m. Express this number in standard form. *(**Answer:** 0.000000000025)*

Assessment 3

Fill in the table by writing the numbers in standard notation or in scientific notation as indicated.

Quantity	Standard Notation	Scientific Notation
1. Distance from Pluto to Sun	5,797,000 km	
2. Diameter of Earth	12,700,000,000 m	
3. Diameter of the Sun		1.39×10^{9} km
4. Approximate width of a human red blood cell	0.0000007 m	
5. Mass of Neutron		1.67×10^{-24} g

What I Have Learned?

Instruction: In your own words, write the steps of writing number in standard notation.

____
____
____
____

Day 4

Lesson 9.4 SOLVING PROBLEMS INVOLVING REAL NUMBERS

What's In?

Evaluate the following real numbers:

Real Numbers	Answer
1) /6/ + /5/	
2) $0.5 \div 10$	
3) $\frac{2}{5} + \frac{3}{4}$	
4) $4 \times 1\frac{2}{3}$	
5) $\frac{1}{2} - \frac{1}{3}$	

Evaluate the following real numbers:

Real Numbers	Answer
6) $\left(\frac{4}{9}\right)\left(\frac{9}{2}\right)$	
7) $\frac{3}{7} + \frac{5}{7}$	
8) $-2 + (-2) + [-(10 + 5)]$	
9) the value of $x + y$ if $x = 5$ and $y = 7$	
10) $10^2 + 4^3$	

[**Answer:** 1) 11; 2) 0.05; 3) $1\frac{3}{20}$; 4) $6\frac{2}{3}$; 5) $\frac{1}{6}$; 6) 2; 7) $1\frac{1}{7}$; 8) – 19; 9) 12; 10) 164]

What's New?

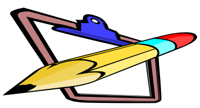
Describe the illustrations below.

A cyclist is traveling 33 kilometers per hour from Baguio City to Tublay, Benguet. How many meters does the cyclist travel in one minute?

What's in it?

This lesson is the culminating part on dealing about real numbers. It combines all the concepts and skills learned in the past lessons on real numbers.

In solving problems involving real numbers, mastery in performing fundamental operations on different properties are needed.



These are the general steps in solving any mathematical problems:

1. Identify the given
2. What do you need to find/problem?
3. What is the equation?
4. Solving proper
5. Conclusion/answer

Example 1: There are 8 packs of fruit juice in a box. How many boxes needed if 40 people are attending the meeting with each receiving 6 packs of fruit juice?

Step 1: What are the given?

a) 8 packs of fruit juice in a box
b) 40 people
c) each person will be receiving 6 packs

Step 2: What is the problem?

How many boxes of fruit juice needed?

Step 3: What is the equation? $(40 \times 6) \div 8$

Step 4: Solving proper:

$40 \times 6 = 240$ packs of fruit juice

$240 \div 8 = 30$ boxes of fruit juice

Step 5: Conclusion:

There are 30 boxes of fruit juice needed

Example 2: Four years ago, Mark's age was half of the age he will be in 10 years. How old is he now?

Step 1: What are the given?

Mark's age 4 years ago	Mark's age in 10 years	Half of the age he will be in 10 years
$x - 4$	$x + 10$	$\frac{1}{2}(x + 10)$

Step 2: What is the problem? Mark's present age

Step 3: What is the equation? $x - 4 = \frac{1}{2}(x + 10)$

Step 4: Solving proper:

Let x be Mark's age now, so;

$x - 4 = \frac{1}{2}(x + 10)$ Equation

$x - 4 = \frac{1}{2}x + 5$ Distributive Property

$x - \frac{1}{2}x = 5 + 4$ Addition Property

$\frac{1}{2}x = 9$ Combine like t

$x = 18$ Multiply both sides by 2 (the reciprocal of $\frac{1}{2}$)

Step 5: Conclusion: Mark's present age is 18 years old.

Example 3: On the last Mathematics quix bee, Lani answered $\frac{4}{9}$ of the problems correctly while Mae answered $\frac{9}{10}$ of the item correctly. If each problem is worth the same amount, who got the higher score?

Step 1: What are the given?

a) Lani answered $\frac{4}{9}$ of the problems correctly

b) Mae answered $\frac{9}{10}$ of the item correctly

Step 2: What is the problem?

A student who got the higher score

Step 3: What is the equation? Lani = $\frac{4}{9}$; Mae = $\frac{9}{10}$;

Step 4: Solving Proper:

Lani		Mae	
$\frac{4}{9}$		$\frac{9}{10}$	Cross Multiply
4(10)	=	9(9)	Simplify
40	<	81	

Step 5: Conclusion: Mae got the higher score than Lani

What's More?

Assessment 4

Solve the following problems.

1. Lucio has 100 coins, a combination of 5 and 1 peso coins, amounting to Php 180.00. How many 1 peso coins does he have?

2. On their previous exam, Rhea answered $\frac{5}{8}$ on the questions correctly and Precious answered $\frac{7}{11}$ of it correctly. If each problem is worth the same amount, who got the higher score?

What I Have Learned?

Instruction: Supply the missing terms in solving problem. Choose the terms inside the box.

Equation	Distributive Property	Addition Property
Combine like terms	Multiply both sides by – 1	

Alex is thrice as old as David. Four years ago, he was 4 times as old as David. How old are they now?

Let Alex = a
David = d

	4 years ago	Present age
Alex	3d – 4	?
David	4(d – 4)	?

1) $3d - 4 = 4(d - 4)$ ____________________
2) $3d - 4 = 4d - 16$ ____________________
3) $3d - 4d = -16 + 4$ ____________________
4) $-d = 12$ ____________________
5) $d = 12$ ____________________

[***Answer:*** *1) Equation; 2) Distributive Property; 3) Addition Property; 4) Combine like terms; 5) Multiply both sides by – 1*]

What I Can Do?

Day 5

Instruction: Read the situations carefully and answer what is being asked. Show your solutions by following the steps in solving mathematical problems.

1. Grace wants to buy a television worth 17,800 pesos. If she buys it using her credit card, she needs to pay 2,222 pesos for 9 months. How much more is the price of the television when purchased on credit card than in cash?

2. Grade 7 – Unity is having an election to decide whether they will go on an educational tour. They will have an educational tour if more than 50% of the class will vote YES. Assume that every member of the class will vote. If 24% of the girls and 18% of the boys will vote YES, will the class go on the educational tour? Explain.

POST ASSESSMENT

Instruction: Let us determine how much you already know about scientific notation and problems involving real numbers. Read each item carefully. Provide the correct answer by writing it on your answer sheet.

1) How many significant digits are there in 2019?
   a) 4 b) 3 c) 2 d) 1

2) Which number has 4 significant digits?
   a) 2072 b) 0.0008 c) 8,000 d) 0.018

3) Which value shows a significant zero - digit?
   a) 0.01 b) 0.002 c) 500 d) 3.04

4) How many significant digits are there in 0.01123?
   a) 1 b) 2 c) 3 d) 4

5) How many significant digits should the product of 1.2 x 3.7?
   a) 1 b) 2 c) 3 d) 4

6) Why is scientific notation important to real numbers?
   a) Because it makes real numbers mathematical
   b) Because it simplifies big and small numbers
   c) Because it helps in mathematical computations
   d) Because it makes use of the powers of 10

7) In the scientific notation 5.0×10^3, which is the coefficient?
   a) 3 b) 5 c) 10 d) x

8) In the scientific notation 1.23×10^{-3}, which shows the number of times the decimal place is moved?
   a) – 3 b) 10 c) 0.23 d) 1.2

9) Which part of the scientific notation indicates the direction of the movement of the decimal point?
   a) Sign of exponent c) Power of 10
   b) Sign of coefficient d) Position of decimal point

1) Which is the scientific notation of 789,000,000,000?
   a) 789×10^{11} c) 0.789×10^{11}
   b) 78.9×10^{11} d) 7.89×10^{11}

11) Which is the scientific notation of 0.000789?
   a) 789×10^{-4} c) 0.789×10^{-4}
   b) 78.9×10^{-4} d) 7.89×10^{-4}

12) Which is the standard form of the scientific notation 4.3×10^5 ?

a) 43,000 b) 430,000 c) 0.00043 d) 0.0043

13) Which is the standard form of the scientific notation 4.3×10^{-5} ?

a) 0.000043 b) 0.00043 c) 0.0043 d) 0.043

14) When a number is written in scientific notation, where is the decimal point located?

a) Decimal point is located between the last two digits of the given number
b) Decimal point is located after the last non-zero digit of the given number.
c) Decimal point is located between any two non-zero digits of the number.
d) Decimal point is located after the first non-zero digit of the given number.

15) When the exponent of 10 in the scientific notation of a number is negative, it means that the number

a) is greater than 1
b) is less than 1.
c) is equal to 1.
d) could not be determined.

16. A deep-freeze compartment is maintained at a temperature of 12°C below zero. If the room temperature is 31°C, how much warmer is the room temperature than the temperature in the deep-freeze compartment.

a) 43°C b) 44°C c) 45°C d) 46°C

17) Yohan can type 50 words per minute, how many words can Yohan type in 20 minutes?

a) 10 words b) 100 words c) 500 words d) 1000 words

18) A hair dryer uses 1.2 kilowatt (KW) of power while a hair iron uses 1.14 kilowatt (KW) of power. How much power is needed if you wish to use the two at the same time?

a) 2.34 KW b) 2.35 KW c) 2.36 KW d) 2.37 KW

19) An elevator is on the ground floor. It goes up 8 floors, then 5 down, and the 4 up. What is its final position from the ground floor?

b) 6^{th} floor b) 7^{th} floor c) 8^{th} floor d) 9^{th} floor

20) Hanna donated $\frac{2}{5}$ of her monthly allowance to the typhoon victims. If her monthly allowance is 3,500 pesos, how much did she donate?

a)1000 pesos b) 1200 pesos c) 1400 pesos d) 1600 pesos

ANSWER KEY

WHAT I KNOW			
1) d	6) c	11) c	16) a
2) d	7) b	12) a	17) c
3) c	8) a	13) a	18) b
4) c	9) c	14) c	19) d
5) c	10) c	15) a	20) c

ASSESSMENT 1
232.86 mL 5 significant digits

ASSESSMENT 2
A R C H I M E D E S

ASSESSMENT 3
1) 5.797×10^{6} km
2) 1.27×10^{10} m
3) 1390000000 km
4) 7×10^{-7} m
5) 0.00000000000000000000000167 g

ASSESSMENT 4
1) 80 pcs of 1 peso coins
2) Precious

POST ASSESSMENT			
1) a	6) b	11) d	16) a
2) a	7) b	12) b	17) d
3) d	8) a	13) a	18) a
4) d	9) a	14) d	19) b
5) c	10) d	15) b	20) c

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